

1 ECR Data-Driven Targeting

1.1 Plant, section and admissible perturbations

The Lorenz system is written in full, since the second equation is what defines the surface of section used throughout,

$$\dot{x} = \sigma(y - x), \quad \dot{y} = \rho x - y - xz, \quad \dot{z} = xy - \beta z, \quad (1)$$

with state $\mathbf{u} = [x \ y \ z]^\top$, flow $\varphi_t(\mathbf{u}; \mathbf{p})$ and control parameters $\mathbf{p} = [\sigma \ \rho \ \beta]^\top \in \mathbb{R}^r$, $r = 3$. The section is the half-plane $\Sigma = \{\mathbf{u} : y = y_{\text{PSS}}, \dot{y} < 0\}$ with $y_{\text{PSS}} = \sqrt{\beta_{\text{nom}}(\rho_{\text{nom}} - 1)} = 8.4853$; transversality requires $\dot{y}|_\Sigma \neq 0$ by (1), and the sign condition selects the branch that excludes the equilibrium C^+ . With the first return time $\tau = \min\{t > 0 : \varphi_t(\mathbf{u}; \mathbf{p}) \in \Sigma\}$ and $y \equiv y_{\text{PSS}}$ on Σ , the observed state at the n -th crossing is $\mathbf{z}_n = [x(t_n) \ z(t_n)]^\top \in \mathbb{R}^N$, $N = 2$, and the dynamics reduce to the discrete map

$$\mathbf{z}_{n+1} = \mathbf{G}(\mathbf{z}_n, \mathbf{p}_n), \quad (2)$$

the parameter being held constant between crossings. Where only a scalar measurement $s(t)$ is available, the delay vector $[s(t_n) \ s(t_n - T) \ \dots]^\top$ replaces the section state and (2) still holds generically.

Each parameter is confined to $\Pi = \{\mathbf{p} : |p^i - p_{\text{nom}}^i| < \delta p_{\text{max}}^i, i = 1, \dots, r\}$, here $\mathbf{p}_{\text{nom}} = [10 \ 28 \ 8/3]^\top$ and $\delta \mathbf{p}_{\text{max}} = [0.30 \ 0.84 \ 0.08]^\top$, i.e. 3% of each nominal value. That budget is a design choice carried over from the source paper, not a property of the method: Π need only be small enough to leave the attractor intact, and both the extent of the control regions and the reaching time scale with how large it is. With the normalised perturbation $\Delta \mathbf{p}_n = (\mathbf{p}_n - \mathbf{p}_{\text{nom}}) \oslash \delta \mathbf{p}_{\text{max}}$, admissibility is the box test $\|\Delta \mathbf{p}_n\|_\infty \leq 1$. The target is a fixed point of the controlled map at nominal parameters, $\mathbf{z}^* = \mathbf{G}(\mathbf{z}^*, \mathbf{p}_{\text{nom}})$ — a period-1 unstable periodic orbit of the flow — located by Newton iteration on $\mathbf{G}(\mathbf{z}, \mathbf{p}_{\text{nom}}) - \mathbf{z}$ with a numerically differentiated Jacobian: $\mathbf{z}^* = [14.2522 \ 39.7867]^\top$, with map eigenvalues $\lambda_1 = 4.71$ and $\lambda_2 \approx 10^{-9}$, so the return map is effectively one-dimensional. The displacement reachable within Π exceeds the target radius $\delta = 0.30$ tenfold, which makes single-step targeting feasible.

1.2 Control regions and their empirical construction

With $\mathcal{B}_\delta = \{\mathbf{z} : \|\mathbf{z} - \mathbf{z}^*\| < \delta\}$, the target region and the extended control regions are

$$T_0 = \left\{ \mathbf{z} \in \mathcal{B}_\delta : \exists \mathbf{p} \in \Pi, \mathbf{G}(\mathbf{z}, \mathbf{p}) \in \mathcal{B}_\delta \right\}, \quad (3)$$

$$T_i = \left\{ \mathbf{z} \notin \bigcup_{j < i} T_j : \exists \mathbf{p} \in \Pi, \mathbf{G}(\mathbf{z}, \mathbf{p}) \in T_{i-1} \right\}, \quad i = 1, \dots, K. \quad (4)$$

The regions are disjoint by construction, membership of T_i certifies capture of T_0 within i admissible steps, and their union is the activation region, outside which $\mathbf{p}_{\text{nom}}$ is applied. Both are existence statements, settled empirically: the plant is run from many initial conditions with an independent $\mathbf{p}_n \sim \mathcal{U}(\Pi)$ applied at each crossing, giving the transitions $\mathcal{D} = \{(\mathbf{z}_n, \mathbf{p}_n, \mathbf{z}_{n+1})\}_{n=1}^M$ in which the applied $\mathbf{p}_n$ is the witness. Level zero follows directly from (3) as $\mathcal{I}_0 = \{n : \|\mathbf{z}_n - \mathbf{z}^*\| < \delta \wedge \|\mathbf{z}_{n+1} - \mathbf{z}^*\| < \delta\}$, and for $i \geq 1$ $\mathcal{I}_i = \{n : \mathcal{M}_{i-1}(\mathbf{z}_{n+1}) = 1 \wedge \mathcal{M}_j(\mathbf{z}_n) = 0 \ \forall j < i\}$, membership of the successor in T_{i-1} being decided by the already-trained level- $(i-1)$ model acting as an oracle $\mathcal{M}_{i-1} : \mathbb{R}^N \rightarrow \{0, 1\}$, defined below. Region i is thus trained on $\{(\mathbf{z}_n, \Delta \mathbf{p}_n)\}_{n \in \mathcal{I}_i}$ — the state, and the perturbation observed to move it one region inward — bottom-up, each level supplying the oracle for the next.

1.3 Clustering, region assignment and local models

ECR-I fits one network over the whole activation region and ECR-II one per region; ECR-III partitions each region first, since data in the outer regions is scattered. On $\mathcal{Z}_i = \{\mathbf{z}_n\}_{n \in \mathcal{I}_i}$ the clusters C_{ij} are the equivalence classes of the transitive closure of $\mathbf{z} \sim \mathbf{z}' \iff \|\mathbf{z} - \mathbf{z}'\| \leq r_i$, with r_i taken at the first minimum of the histogram of pairwise distances. Each cluster carries a mean $\boldsymbol{\mu}_{ij}$ and a ridge-regularised covariance $\boldsymbol{\Sigma}_{ij}$, and a state is assigned to the cluster minimising the normalised Mahalanobis distance

$$\text{NMD}_{ij}^2(\mathbf{z}_n) = (\mathbf{z}_n - \boldsymbol{\mu}_{ij})^\top \underline{\mathbf{N}}_{ij} (\mathbf{z}_n - \boldsymbol{\mu}_{ij}), \quad \underline{\mathbf{N}}_{ij} = \left(\boldsymbol{\Sigma}_{ij} / \det(\boldsymbol{\Sigma}_{ij})^{1/N} \right)^{-1}, \quad (5)$$

the unit-determinant normalisation removing the size of a cluster but not its shape, so clusters of different populations compete on equal terms (plain, variance- and trace-normalised readings are alternatives). Each cluster also carries a Gaussian RBF network mapping the standardised state $\tilde{\mathbf{z}}$ to a normalised perturbation, $\widehat{\Delta \mathbf{p}}(\mathbf{z}) = \mathbf{W}^\top [\phi_1 \cdots \phi_{n_c} 1]^\top$ with $\phi_k = \exp(-\|\tilde{\mathbf{z}} - \mathbf{c}_k\|^2 / 2s_k^2)$, centres from k -means and weights $\mathbf{W} = (\boldsymbol{\Phi}^\top \boldsymbol{\Phi} + \lambda \mathbf{I})^{-1} \boldsymbol{\Phi}^\top \mathbf{Y}$ in closed form. The oracle used above is an admissibility test on the nearest cluster's network, $\mathcal{M}_i(\mathbf{z}) = \mathbb{1}[\|\widehat{\Delta \mathbf{p}}^{(ij^*)}(\mathbf{z})\|_\infty \leq 1]$ with $j^* = \arg \min_j \text{NMD}_{ij}(\mathbf{z})$, optionally gated on the cluster's own training NMD, since (5) returns a nearest cluster even for states far outside the data.

1.4 Control law and performance measures

At each crossing the state is measured, a region is identified and the corresponding perturbation applied,

$$\mathbf{p}_n = \begin{cases} \mathbf{p}_{\text{nom}} + \widehat{\Delta \mathbf{p}}^{(i^* j^*)}(\mathbf{z}_n) \odot \delta \mathbf{p}_{\text{max}}, & \text{if an admissible } (i^*, j^*) \text{ exists,} \\ \mathbf{p}_{\text{nom}}, & \text{otherwise,} \end{cases} \quad (6)$$

so the applied perturbation is admissible by construction. The variants differ only in how the pair is found: ECR-II takes the first admissible answer in ascending region order, $i^* = \min\{i : \|\widehat{\Delta \mathbf{p}}^{(i)}(\mathbf{z}_n)\|_\infty \leq 1\}$, whereas ECR-III obtains it analytically as $(i^*, j^*) = \arg \min_{i,j} \text{NMD}_{ij}(\mathbf{z}_n)$ over the clusters of all regions at once, trading a sequential search over networks for a distance computation. Performance is measured by the reaching time and its mean over initial conditions on the attractor,

$$n_r(\mathbf{z}_0) = \min\{n \geq 0 : \|\mathbf{z}_n - \mathbf{z}^*\| < \delta\}, \quad \bar{n}_r = \frac{1}{M_{\text{ic}}} \sum_{m=1}^{M_{\text{ic}}} \min\{n_r(\mathbf{z}_0^{(m)}), n_{\text{max}}\}, \quad (7)$$

censored at a horizon n_{max} , together with the capture rate $\Pr\{n_r < n_{\text{max}}\}$ and the retention — the fraction of steps after capture spent inside $\mathcal{B}_\delta$ — which separates targeting from stabilisation. Without targeting $\bar{n}_r$ is the natural recurrence time to $\mathcal{B}_\delta$, some 15 steps here.